Mathematical Analysis II
Week 5 Homework (April 2)

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Section 9.3

4. 试求由下列方程所确定的隐函数的微分.

(1) $\cos^2 x + \cos^2 y + \cos^2 z = 1$, 求 $\mathrm{d}z$;

解. 两边求微分，得

$$-2\sin x\cos x\,\mathrm{d}x - 2\sin y\cos y\,\mathrm{d}y - 2\sin z\cos z\,\mathrm{d}z = 0$$

因为 $\sin\alpha\cos\alpha = \sin(2\alpha)$，因此 $-\sin 2x\,\mathrm{d}x - \sin 2y\,\mathrm{d}y = \sin 2z\,\mathrm{d}z$

从而

$$\mathrm{d}z = -\frac{\sin 2x}{\sin 2z}\,\mathrm{d}x - \frac{\sin 2y}{\sin 2z}\,\mathrm{d}y$$

(2) $xyz = x + y + z$, 求 $\mathrm{d}z$;

解. 两边求微分，得

$$xy\,\mathrm{d}z + xz\,\mathrm{d}y + yz\,\mathrm{d}x = \mathrm{d}x + \mathrm{d}y + \mathrm{d}z$$

因此 $(xz-1)\,\mathrm{d}y + (yz-1)\,\mathrm{d}x = (1-xy)\,\mathrm{d}z$

从而

$$\mathrm{d}z = \frac{yz-1}{1-xy}\,\mathrm{d}x + \frac{xz-1}{1-xy}\,\mathrm{d}y$$

(3) $u^3 - 3(x+y)u^2 + z^3 = 0$, 求 $\mathrm{d}u$;

解. 左边 $= u^3 - 3xu^2 - 3yu^2 + z^3$

两边求微分，得

$$3u^2\,\mathrm{d}u - 3u^2\,\mathrm{d}x - 3u^2\,\mathrm{d}y - 6xu\,\mathrm{d}u - 6yu\,\mathrm{d}u + 3z^2\,\mathrm{d}z = 0$$

因此 $3u(u-2x-2y)\,\mathrm{d}u = 3u^2(\mathrm{d}x + \mathrm{d}y) - 3z^2\,\mathrm{d}z$

从而

$$\mathrm{d}u = \frac{u}{u-2x-2y}\,\mathrm{d}x + \frac{u}{u-2x-2y}\,\mathrm{d}y - \frac{z^2}{u^2-2ux-2uy}\,\mathrm{d}z$$

(4) $F(x-y, y-z, z-x)=0$, 求 $\mathrm{d}z$.

解. 两边求微分，得

$$F_1(\mathrm{d}x-\mathrm{d}y)+F_2(\mathrm{d}y-\mathrm{d}z)+F_3(\mathrm{d}z-\mathrm{d}x)=0$$

因此 $(F_3-F_2)\,\mathrm{d}z=(F_3-F_1)\,\mathrm{d}x+(F_1-F_2)\,\mathrm{d}y$

从而

$$\mathrm{d}z=\frac{F_3-F_1}{F_3-F_2}\,\mathrm{d}x+\frac{F_1-F_2}{F_3-F_2}\,\mathrm{d}y$$

7. 设 $z=z(x,y)$ 是由方程 $\varphi(cx-az, cy-bz)=0$ 所确定的隐函数，试证：不论 φ 为怎样的可微函数，都有

$$a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}=c.$$

证. 对 x 求偏导，得

$$\varphi_1\left(c-a\frac{\partial z}{\partial x}\right)+\varphi_2\left(-b\frac{\partial z}{\partial x}\right)=0$$

因此 $c\varphi_1=(a\varphi_1+b\varphi_2)\frac{\partial z}{\partial x}$

对 y 求偏导，得

$$\varphi_1\left(-a\frac{\partial z}{\partial y}\right)+\varphi_2\left(c-b\frac{\partial z}{\partial y}\right)=0$$

因此 $c\varphi_2=(a\varphi_1+b\varphi_2)\frac{\partial z}{\partial y}$

从而 $a\varphi_1c+b\varphi_2c=\left(a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}\right)(a\varphi_1+b\varphi_2)$

因此

$$a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}=c$$

Q.E.D.

10. 设 $x=x(z)$, $y=y(z)$ 是方程组

$$\begin{cases}x+y+z=0,\\ x^2+y^2+z^2=1\end{cases}$$

所确定的隐函数组，求 $\dfrac{\mathrm{d}x}{\mathrm{d}z}, \dfrac{\mathrm{d}y}{\mathrm{d}z}$;

解. 求微分，得

$$\begin{cases}\mathrm{d}x+\mathrm{d}y+\mathrm{d}z=0\\ 2x\,\mathrm{d}x+2y\,\mathrm{d}y+2z\,\mathrm{d}z=0\end{cases}$$

因此 $\mathrm{d}y=-\mathrm{d}x-\mathrm{d}z$，代入得 $(x-y)\,\mathrm{d}x+(z-y)\,\mathrm{d}z=0$，从而

$$\frac{\mathrm{d}x}{\mathrm{d}z}=-\frac{z-y}{x-y}$$

同理可得

$$\frac{\mathrm{d}y}{\mathrm{d}z}=-\frac{z-x}{y-x}$$

11. 设 $u=u(x,y)$, $v=v(x,y)$ 是由下列方程组所确定的隐函数组，求 $\dfrac{\partial(u,v)}{\partial(x,y)}$.

(1)

$$\begin{cases} u^2+v^2+x^2+y^2=1, \\ u+v+x+y=0; \end{cases}$$

解. 对 x 求偏导，得

$$\begin{cases} 2u\frac{\partial u}{\partial x}+2v\frac{\partial v}{\partial x}+2x=0 \\ \frac{\partial u}{\partial x}+\frac{\partial v}{\partial x}+1=0 \end{cases}$$

解得

$$\begin{cases} \dfrac{\partial u}{\partial x}=\dfrac{v-x}{u-v} \\ \dfrac{\partial v}{\partial x}=\dfrac{u-x}{v-u} \end{cases}$$

同理可得

$$\begin{cases} \dfrac{\partial u}{\partial y}=\dfrac{v-y}{u-v} \\ \dfrac{\partial v}{\partial y}=\dfrac{u-y}{v-u} \end{cases}$$

因此

$$\frac{\partial(u,v)}{\partial(x,y)}=\begin{pmatrix} \dfrac{v-x}{u-v} & \dfrac{v-y}{u-v} \\ \dfrac{u-x}{v-u} & \dfrac{u-y}{v-u} \end{pmatrix}$$

(2)

$$\begin{cases} xu-yv=0, \\ yu+xv=1; \end{cases}$$

解. 对 x 求偏导，得

$$\begin{cases} u+x\frac{\partial u}{\partial x}-y\frac{\partial v}{\partial x}=0 \\ y\frac{\partial u}{\partial x}+v+x\frac{\partial v}{\partial x}=0 \end{cases}$$

解得

$$\begin{cases} \dfrac{\partial u}{\partial x}=-\dfrac{ux+vy}{x^2+y^2} \\ \dfrac{\partial v}{\partial x}=\dfrac{uy-vx}{x^2+y^2} \end{cases}$$

对 y 求偏导，得

$$\begin{cases} x\frac{\partial u}{\partial y}-v-y\frac{\partial v}{\partial y}=0 \\ u+y\frac{\partial u}{\partial y}+x\frac{\partial v}{\partial y}=0 \end{cases}$$

解得

$$\begin{cases}\dfrac{\partial u}{\partial y}=\dfrac{vx-uy}{x^2+y^2}\\ \dfrac{\partial v}{\partial y}=-\dfrac{ux+vy}{x^2+y^2}\end{cases}$$

因此

$$\frac{\partial(u,v)}{\partial(x,y)}=\begin{pmatrix}-\dfrac{ux+vy}{x^2+y^2} & \dfrac{vx-uy}{x^2+y^2}\\ \dfrac{uy-vx}{x^2+y^2} & -\dfrac{ux+vy}{x^2+y^2}\end{pmatrix}$$

(3)

$$\begin{cases}u=f(ux,v+y),\\ v=g(u-x,v^2y).\end{cases}$$

解. 对 x 求偏导，得

$$\begin{cases}\dfrac{\partial u}{\partial x}=f_1\left(x\frac{\partial u}{\partial x}+u\right)+f_2\frac{\partial v}{\partial x}\\ \dfrac{\partial v}{\partial x}=g_1\left(\frac{\partial u}{\partial x}-1\right)+g_2\left(2vy\frac{\partial v}{\partial x}\right)\end{cases}$$

整理得

$$\begin{cases}(1-f_1x)\frac{\partial u}{\partial x}-f_2\frac{\partial v}{\partial x}=f_1u\\ -g_1\frac{\partial u}{\partial x}+(1-2g_2vy)\frac{\partial v}{\partial x}=-g_1\end{cases}$$

记 $J=(1-f_1x)(1-2g_2vy)-f_2g_1$，则

$$\begin{cases}\dfrac{\partial u}{\partial x}=\dfrac{\begin{vmatrix}uf_1 & -f_2\\ -g_1 & 1-2g_2vy\end{vmatrix}}{J}\\ \dfrac{\partial v}{\partial x}=\dfrac{\begin{vmatrix}1-f_1x & uf_1\\ -g_1 & -g_1\end{vmatrix}}{J}\end{cases}$$

对 y 求偏导，得

$$\begin{cases}\dfrac{\partial u}{\partial y}=f_1x\frac{\partial u}{\partial y}+f_2\left(\frac{\partial v}{\partial y}+1\right)\\ \dfrac{\partial v}{\partial y}=g_1\frac{\partial u}{\partial y}+g_2\left(2vy\frac{\partial v}{\partial y}+v^2\right)\end{cases}$$

整理得

$$\begin{cases}(f_1x-1)\frac{\partial u}{\partial y}+f_2\frac{\partial v}{\partial y}=-f_2\\ g_1\frac{\partial u}{\partial y}+(2vyg_2-1)\frac{\partial v}{\partial y}=-v^2g_2\end{cases}$$

因此

$$\begin{cases} \dfrac{\partial u}{\partial y} = \dfrac{\begin{vmatrix} -f_2 & f_2 \\ -v^2 g_2 & 2vyg_2 - 1 \end{vmatrix}}{J} \\ \dfrac{\partial v}{\partial y} = \dfrac{\begin{vmatrix} f_1 x - 1 & -f_2 \\ g_1 & -v^2 g_2 \end{vmatrix}}{J} \end{cases}$$

从而

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{pmatrix} \dfrac{\begin{vmatrix} uf_1 & -f_2 \\ -g_1 & 1-2g_2vy \end{vmatrix}}{(1-f_1x)(1-2g_2vy)-f_2g_1} & \dfrac{\begin{vmatrix} -f_2 & f_2 \\ -v^2g_2 & 2vyg_2-1 \end{vmatrix}}{(1-f_1x)(1-2g_2vy)-f_2g_1} \\ \dfrac{\begin{vmatrix} 1-f_1x & uf_1 \\ -g_1 & -g_1 \end{vmatrix}}{(1-f_1x)(1-2g_2vy)-f_2g_1} & \dfrac{\begin{vmatrix} f_1x-1 & -f_2 \\ g_1 & -v^2g_2 \end{vmatrix}}{(1-f_1x)(1-2g_2vy)-f_2g_1} \end{pmatrix}$$

15. 设 $u = u(x,y)$, $v = v(x,y)$ 是由方程 $F(x,y,u,v) = 0$ 和 $G(x,y,u,v) = 0$ 所确定的隐函数，其中 F 和 G 都具有一阶连续偏导数。求 $\mathrm{d}u$, $\mathrm{d}v$.

解. 两边求微分，得

$$\begin{cases} F_3 \,\mathrm{d}u + F_4 \,\mathrm{d}v = -F_1 \,\mathrm{d}x - F_2 \,\mathrm{d}y \\ G_3 \,\mathrm{d}u + G_4 \,\mathrm{d}v = -G_1 \,\mathrm{d}x - G_2 \,\mathrm{d}y \end{cases}$$

解得

$$\begin{cases} \mathrm{d}u = \dfrac{\begin{vmatrix} -F_1 \,\mathrm{d}x - F_2 \,\mathrm{d}y & F_4 \\ -G_1 \,\mathrm{d}x - G_2 \,\mathrm{d}y & G_4 \end{vmatrix}}{\begin{vmatrix} F_3 & F_4 \\ G_3 & G_4 \end{vmatrix}} \\ \mathrm{d}v = \dfrac{\begin{vmatrix} F_3 & -F_1 \,\mathrm{d}x - F_2 \,\mathrm{d}y \\ G_3 & -G_1 \,\mathrm{d}x - G_2 \,\mathrm{d}y \end{vmatrix}}{\begin{vmatrix} F_3 & F_4 \\ G_3 & G_4 \end{vmatrix}} \end{cases}$$

16. 函数 $u = u(x,y)$ 由方程组 $u = f(x,y,z,t)$, $g(y,z,t) = 0$, $h(z,t) = 0$ 定义，求 $\dfrac{\partial u}{\partial x}$, $\dfrac{\partial u}{\partial y}$.

解. 对 x 求偏导，得

$$\begin{cases} \dfrac{\partial u}{\partial x} = \dfrac{\partial f}{\partial x} + \dfrac{\partial f}{\partial z}\dfrac{\partial z}{\partial x} + \dfrac{\partial f}{\partial t}\dfrac{\partial t}{\partial x} \\ \dfrac{\partial g}{\partial z}\dfrac{\partial z}{\partial x} + \dfrac{\partial g}{\partial t}\dfrac{\partial t}{\partial x} = 0 \\ \dfrac{\partial h}{\partial z}\dfrac{\partial z}{\partial x} + \dfrac{\partial h}{\partial t}\dfrac{\partial t}{\partial x} = 0 \end{cases}$$

因此

$$\begin{cases} \dfrac{\partial z}{\partial x} = 0 \\ \dfrac{\partial t}{\partial x} = 0 \end{cases}$$

从而

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x}$$

对 y 求偏导，得

$$\begin{cases} \dfrac{\partial u}{\partial y} = \dfrac{\partial f}{\partial y} + \dfrac{\partial f}{\partial z}\dfrac{\partial z}{\partial y} + \dfrac{\partial f}{\partial t}\dfrac{\partial t}{\partial y} \\ \dfrac{\partial g}{\partial y} + \dfrac{\partial g}{\partial z}\dfrac{\partial z}{\partial y} + \dfrac{\partial g}{\partial t}\dfrac{\partial t}{\partial y} = 0 \\ \dfrac{\partial h}{\partial z}\dfrac{\partial z}{\partial y} + \dfrac{\partial h}{\partial t}\dfrac{\partial t}{\partial y} = 0 \end{cases}$$

整理得

$$\begin{cases} \dfrac{\partial g}{\partial z}\dfrac{\partial z}{\partial y} + \dfrac{\partial g}{\partial t}\dfrac{\partial t}{\partial y} = -\dfrac{\partial g}{\partial y} \\ \dfrac{\partial h}{\partial z}\dfrac{\partial z}{\partial y} + \dfrac{\partial h}{\partial t}\dfrac{\partial t}{\partial y} = 0 \end{cases}$$

解得

$$\begin{cases} \dfrac{\partial z}{\partial y} = -\dfrac{\dfrac{\partial g}{\partial y}\dfrac{\partial h}{\partial t}}{\begin{vmatrix} \dfrac{\partial g}{\partial z} & \dfrac{\partial g}{\partial t} \\ \dfrac{\partial h}{\partial z} & \dfrac{\partial h}{\partial t} \end{vmatrix}} \\ \dfrac{\partial t}{\partial y} = \dfrac{\dfrac{\partial g}{\partial y}\dfrac{\partial h}{\partial z}}{\begin{vmatrix} \dfrac{\partial g}{\partial z} & \dfrac{\partial g}{\partial t} \\ \dfrac{\partial h}{\partial z} & \dfrac{\partial h}{\partial t} \end{vmatrix}} \end{cases}$$

因此

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial y} + \frac{\dfrac{\partial g}{\partial y}\left(\dfrac{\partial f}{\partial t}\dfrac{\partial h}{\partial z} - \dfrac{\partial f}{\partial z}\dfrac{\partial h}{\partial t}\right)}{\dfrac{\partial g}{\partial z}\dfrac{\partial h}{\partial t} - \dfrac{\partial g}{\partial t}\dfrac{\partial h}{\partial z}}$$

2

Let $\mathrm{GL}(n,\mathbb{R}) \subset \mathbb{R}^{n^2}$ be the set of $n \times n$ invertible matrices (by identifying $A = (x_{ij})$ with the vector $(x_{11}, \ldots, x_{1n}, x_{21}, \ldots, x_{nn})^T$).

(a) Using the map $\det : \mathbb{R}^{n^2} \to \mathbb{R}$ to prove $\mathrm{GL}(n,\mathbb{R})$ is open. As a consequence, the small perturbation of an invertible matrix is also invertible.

证. 由于 A 可逆，因此 $\det(A) \neq 0$。

从而 $\mathrm{GL}(n,\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$。显然 $\mathbb{R} \setminus \{0\}$ 是开集。

由于 $\det:\mathbb{R}^{n^2}\to\mathbb{R}$ 连续，因此 $\mathrm{GL}(n,\mathbb{R})$ 是开集。

从而可逆矩阵的任意充分小扰动仍可逆。

Q.E.D.

(b) Show the map $\varphi:\mathrm{GL}(n,\mathbb{R})\to\mathrm{GL}(n,\mathbb{R})$ sending A to A^{-1} is continuous.

证.

$$A^{-1}=\frac{\mathrm{adj}(A)}{\det(A)}$$

由于 A 的伴随矩阵 $\mathrm{adj}(A)$ 中的每个元素均是关于 x_{ij} 的多项式函数，因此 adj 是连续函数。

又由于 $\det(A)\neq 0$ 且连续，从而 φ 是连续函数。

Q.E.D.

(c) Show that φ is differentiable, and find $\mathrm{d}\varphi|_A$.

证. 定义映射

$$F:\mathrm{GL}(n,\mathbb{R})\times\mathbb{R}^{n^2}\to\mathbb{R}^{n^2}$$
$$F(A,B)\mapsto AB-I$$

由于矩阵乘法、加法连续，因此 $F\in C^1$。

取 $B=A^{-1}$，则 $F(A,B)=AB-I=0$。

固定 A，将 $B\mapsto AB-I$ 视为关于 B 的线性映射，则对任意 $H\in\mathbb{R}^{n^2}$，有

$$\left.\frac{\partial F}{\partial B}\right|_{(A,B)}(H)=AH$$

由于 A 可逆，因此线性映射 $H\mapsto AH$ 可逆。

由隐函数定理，存在 A 的邻域使得方程 $F(A,B)=0$ 唯一确定隐函数 $B=\varphi(A)=A^{-1}$，且 φ 在该邻域上可微。

对 $A\varphi(A)\equiv I$ 两边取微分，得

$$\mathrm{d}A\cdot\varphi(A)+A\mathrm{d}\varphi(A)=0$$

记 $\mathrm{d}A=H$，$\varphi(A)=B=A^{-1}$，代入得

$$HA^{-1}+A\,\mathrm{d}\varphi|_A\,(H)=0$$

从而

$$\mathrm{d}\varphi|_A\,(H)=-A^{-1}HA^{-1},\quad H\in\mathbb{R}^{n^2}$$